SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10001	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Hari Prakash. N	**Reg. No.:**	192572038
Date:	13/10/2026	**Duration:**	60 minutes

Code of Conduct

I Hari Prakash. N / 192572038 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Hari Prakash. N

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices.

Agile Sprint Planning & User Story Workshop

Problem Statement: Food Waste Management and Donation Platform

1. Project Vision and Stakeholders:

Project Vision:

To provide a coordinated platform that helps food donors share details of safe surplus food and enables eligible recipient organizations to identify, reserve, collect, and distribute it before it becomes unsuitable. The platform aims to reduce avoidable food waste, improve access to available food, and make donation activities transparent and traceable.

Objectives:

• Allow restaurants, hotels, grocery stores, event organizers, and individuals to list surplus food with quantity, food type, preparation or expiry time, location, and collection window.
• Help charities, food banks, shelters, and community kitchens find suitable donations based on location, availability, food requirements, and capacity.
• Support reservation, collection or delivery coordination, and status tracking from listing to completed donation.
• Provide clear information about food suitability and time limits, while allowing authorized users to update donation status.
• Maintain records that help users review completed, cancelled, or expired donations and understand the platform's impact.

Key Stakeholders:

Stakeholder	**Role / Interest**
Food donors (restaurants, shops, event organizers, individuals)	List surplus food accurately and update availability, quantity, and collection details.
Recipient organizations (NGOs, shelters, food banks, community kitchens)	Search for suitable donations, reserve them, and confirm collection or receipt.
Volunteer / delivery partners	Coordinate pickup and delivery when transport assistance is needed.
Platform administrator	Manage accounts, review reports, handle complaints, and monitor platform activity.
Beneficiaries / community members	Benefit indirectly when suitable donated food reaches recipient organizations.
Food-safety / local authorities	Provide relevant food-handling guidance and requirements where applicable.
Development and testing team	Gather requirements, design workflows, build, test, and maintain the platform.

2. Epics and User Stories:

The following epics group the main requirements. User stories use the required Agile format.

Epic	User Story
E1 – Account and Profile Management	US1: As a donor, I want to register and manage my profile, so that recipients can identify a reliable donation source.
E2 – Surplus Food Listing	US2: As a donor, I want to list surplus food with its type, quantity, preparation/expiry details, location, and collection window, so that recipients can judge whether it is suitable.
E3 – Donation Discovery and Matching	US3: As a recipient organization, I want to search and filter available donations by location, food type, quantity, and time, so that I can find food that meets our needs.
E4 – Reservation and Coordination	US4: As a recipient organization, I want to reserve an available listing and coordinate collection, so that the donation is not claimed by multiple recipients.
E5 – Donation Status Tracking	US5: As a donor or recipient, I want to update and view donation status, so that both sides know whether the food is available, reserved, collected, completed, cancelled, or expired.
E6 – Administration and Reporting	US6: As an administrator, I want to review user activity and donation records, so that I can resolve issues and monitor platform usage.

3. Acceptance Criteria:

Acceptance criteria are written in Given–When–Then form and can be checked during testing.

Story	Acceptance Criteria (Given – When – Then)
US1 – Register profile	Given a user has not registered, when they submit all required valid details, then an account is created and a confirmation is shown. Given

	required fields are missing or invalid, when the form is submitted, then the system identifies the fields that need correction.
US2 – Create food listing	Given a donor is signed in, when they enter food type, quantity, preparation/expiry details, location, and collection window and submit the listing, then the listing appears as available. If required details are missing or the collection time is invalid, then the listing is not published and an explanation is shown.
US3 – Search donations	Given available listings exist, when a recipient applies search filters, then only listings matching the selected criteria are displayed. If no listing matches, then the platform displays a clear no-results message.
US4 – Reserve donation	Given a listing is available, when an eligible recipient reserves it, then its status changes to reserved and other recipients cannot reserve the same listing. If it is already reserved or expired, then the reservation is rejected with a message.
US5 – Update status	Given a donation has been reserved, when an authorized donor or recipient updates its progress, then the new status and update time are recorded and visible to relevant users. Unauthorized users cannot change the status.
US6 – View records	Given an administrator is authenticated, when they open donation records and apply filters, then matching records and their statuses are displayed. Non-administrator users cannot access the administrative view.

4. Product Backlog Prioritization:

Priority is based on the platform's core value, user needs, and dependencies. P1 is highest priority.

Priority	Backlog Item	Reason / Dependency
P1	US1 – Register profile	Required to identify users and control access to donor and recipient functions.
P1	US2 – Create food listing	Core function; recipients need accurate available-food information.
P1	US3 – Search donations	Core recipient function; depends on food listings being available.
P1	US4 – Reserve donation	Prevents duplicate claims and enables coordination; depends on listing and recipient functions.
P1	US5 – Update status	Provides traceability for reservation, collection, and completion.
P2	US6 – Administrative records	Supports monitoring and issue resolution after core donation flow is working.
P3	Future enhancements: notifications, route support, analytics	Useful improvements after the minimum donation flow is validated.

5. Sprint Backlog – Sprint 1:

Sprint duration: 2 weeks (assumption). The first sprint focuses on a minimum usable flow: donors can publish surplus food and recipient organizations can find and reserve a suitable listing.

Selected Story	Development / Work Tasks
US1 – Register profile	Confirm required profile fields; design registration and login screens; implement basic input validation; test valid and invalid submissions.
US2 – Create food listing	Define food-listing fields; design listing form; implement required-field and time validation; save and display available listings; test listing creation.
US3 – Search donations	Define search/filter fields; implement filters for location, food type, quantity, and time; show matching listings; test matching and no-result cases.
US4 – Reserve donation	Define reservation and status rules; implement reservation action; prevent duplicate reservations; display reservation confirmation; test competing reservations.

6. Story Point Estimation:

Estimation technique: Planning Poker using the Fibonacci sequence (1, 2, 3, 5, 8, 13). Estimates represent relative effort, complexity, and uncertainty; they are planning estimates, not hours.

Story	Points	Justification
US1 – Register profile	3	Small workflow with basic forms and validation.
US2 – Create food listing	5	Several required fields, time validation, and listing storage/display.
US3 – Search donations	5	Multiple filters and result/no-result behavior need implementation and testing.
US4 – Reserve donation	8	Requires status transitions and protection against duplicate reservations.
US5 – Update status	5	Requires authorized updates and a reliable status history; planned for a later sprint if capacity is limited.
US6 – Administrative records	5	Requires access restrictions and filtered record views; planned for a later sprint.

Sprint 1 planned commitment: US1–US4 = 21 story points. This is a provisional estimate for a new team and should be adjusted after the team's first sprint based on actual capacity.

7. Sprint Goal and Expected Deliverables:

Sprint Goal:

Enable a donor to create an account and publish a surplus-food listing, and enable a recipient organization to find and reserve that listing through a basic, testable donation workflow.

Expected Deliverables:

• Basic donor and recipient registration/profile flow with input validation.

• Surplus-food listing form containing food type, quantity, preparation/expiry details, location, and collection window.

• A page or view showing available food listings.

• Search and filter capability for recipient organizations.

• Reservation function that prevents a listing from being reserved more than once.

• Test cases and recorded results for valid listing, missing fields, filtered search, no matching results, and duplicate reservation attempts.

• A short sprint review demonstration and a list of unresolved issues or improvements for the next sprint.

Assumptions and Limitations

• This sprint plan is an academic planning exercise; the estimates and two-week duration are assumptions and may change after team discussion.

• The platform should encourage donors to provide accurate food information, but a listing alone cannot guarantee food safety. Applicable food-handling rules and recipient checks remain necessary.

• Live GPS routing, payment, automated notifications, and advanced analytics are outside the first sprint and can be considered later.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.

Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	**Mark**
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25